What should be your sample size?

(Intro to effect size and power)

HADC! (HARPER ADAMS DATA CLUB)

Ed Harris

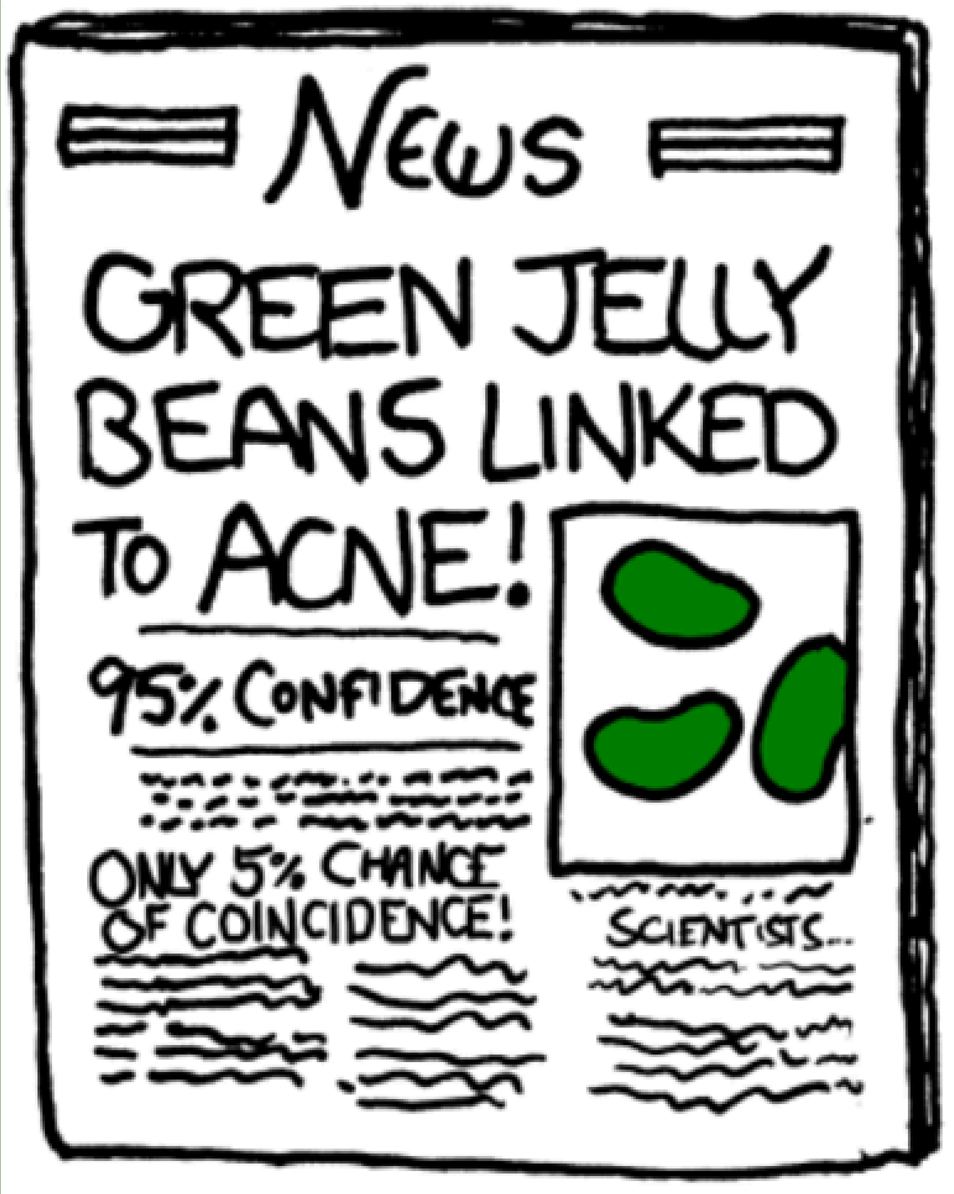
2025-11-18

Effect size thinking

- How much data do you NEED to collect?
- How big is the difference you predict? (be specific)
- Justify your sample size!

First meeting for the Harper Adams R Users Group (HARUp!...?)

Ed Harris
2019.10.16



Effect size thinking and power analysis
(Plus some other HARUp! business)

Statistical power

		Real World	
		Null true	Null false
Conclusion of significance test	Null true	Correct decision	β Type II error (false negative)
	Null false	α Type I error (false positive)	Correct decision

Statistical power

α

- **Type I error rate** is controlled by the researcher.
- **This is the probability cut-off** in a significance test (i.e., 0.05).
- **In principle, any probability value could be chosen** for making the accept/reject decision!

Statistical power

β

- Type II error is also controlled by the researcher.
- To control Type II error, design your experiment to have good statistical power (the good news is that this is easy)
- Power = 1 - beta, in other words the probability you will correctly reject the null hypothesis when the null is false

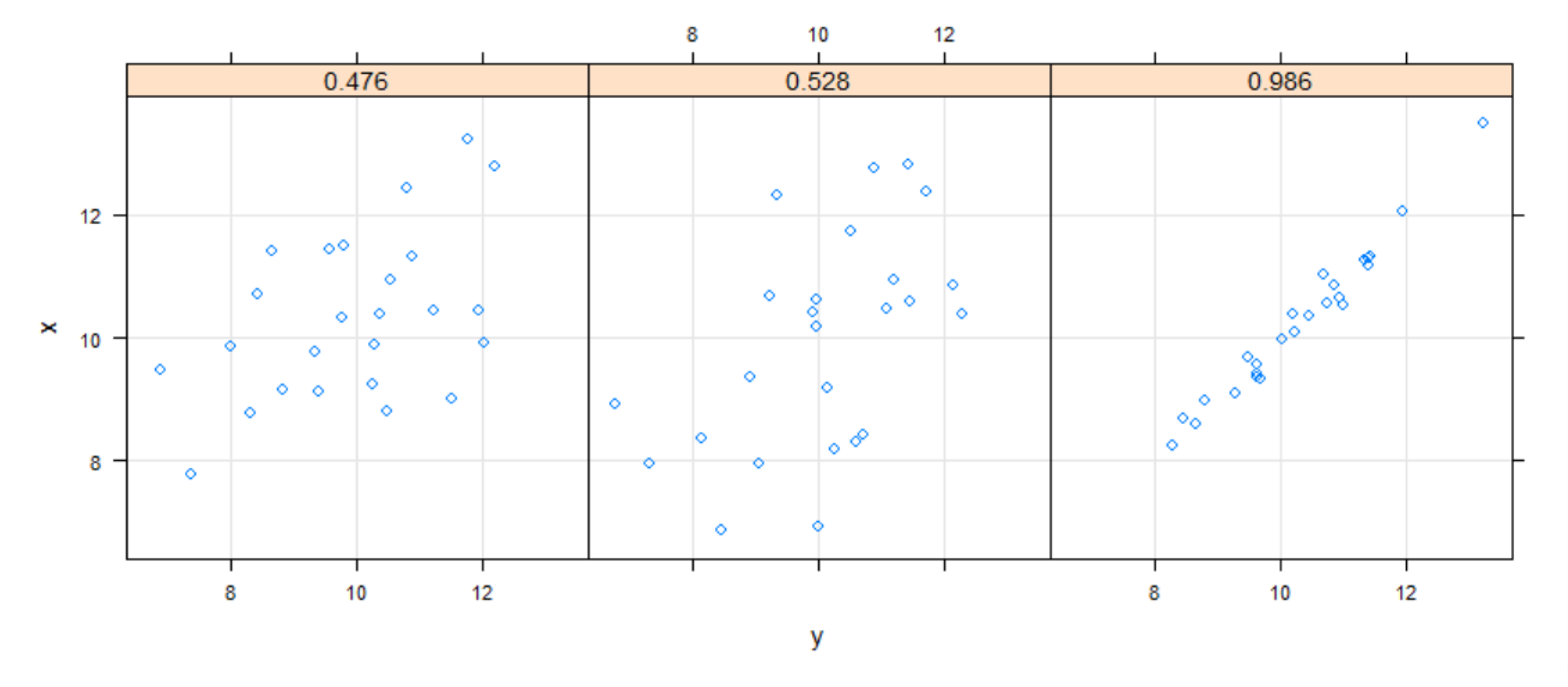
Are you obsessed with power (you should be)?

- **Efficiency**: Research is expensive and time consuming
- **Ethics**: Minimize required sample subjects and maximize their sacrifice
- **Practicality**: With good reason many grant funding agencies now either require or prefer a formal power analysis
- To be blunt, you should probably just **go home if you engage in data collection without conducting a power analysis**

Effect size, there are several types

Correlation

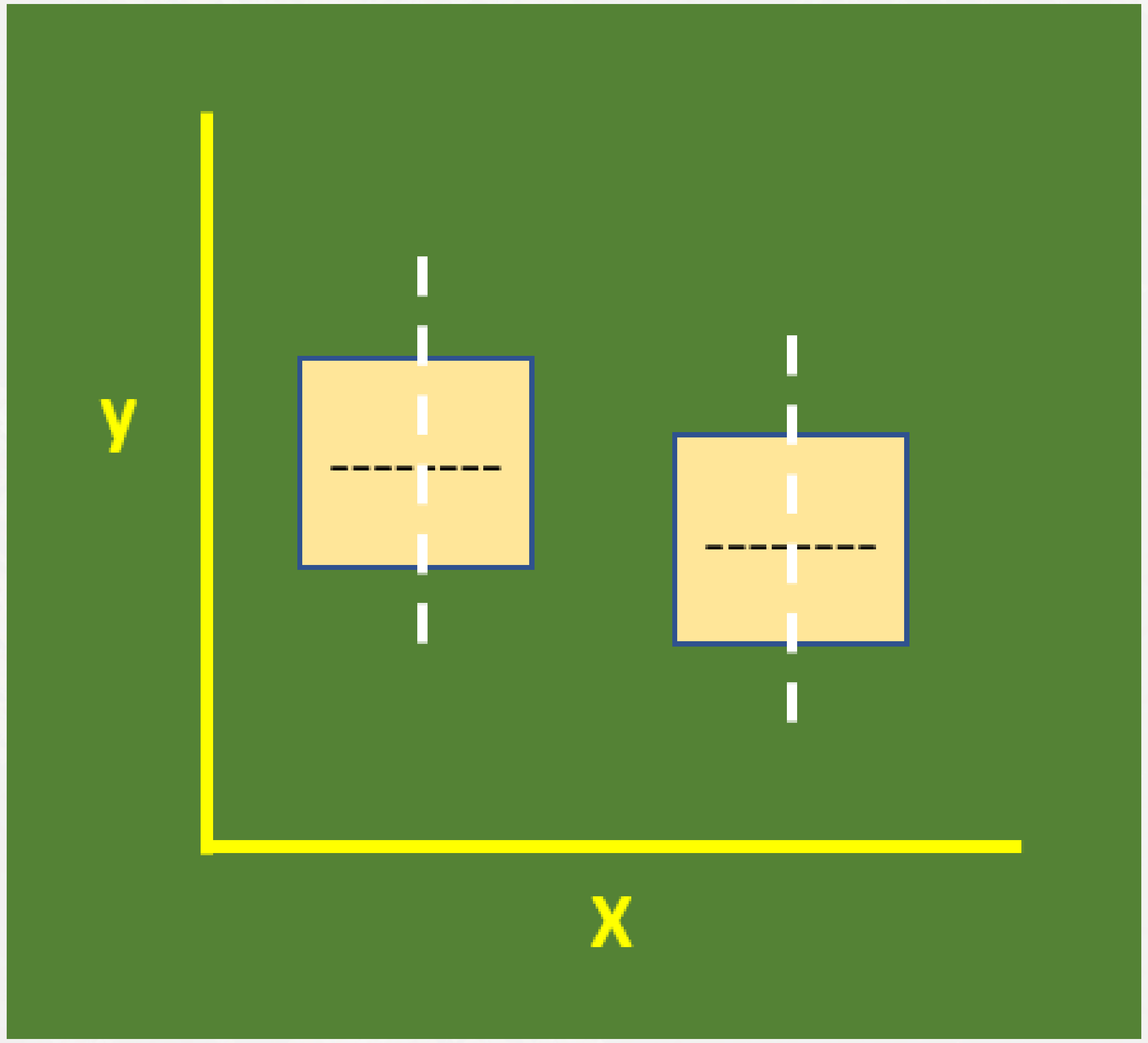
- Measured by the correlation coefficient, r



Effect size, there are two types

Difference

- **Measured by Cohen's d**



For a t-test -> Cohen's d

Cohen's d = **(mean1 - mean2)/(Pooled std dev)**

Literature

QUANTITATIVE METHODS IN PSYCHOLOGY

A Power Primer

Jacob Cohen
New York University

Behavioral Ecology Vol. 15 No. 6: 1044–1045
doi:10.1093/beheco/arh107
Advance Access publication on June 30, 2004

Forum

A farewell to Bonferroni: the problems of low statistical power and publication bias

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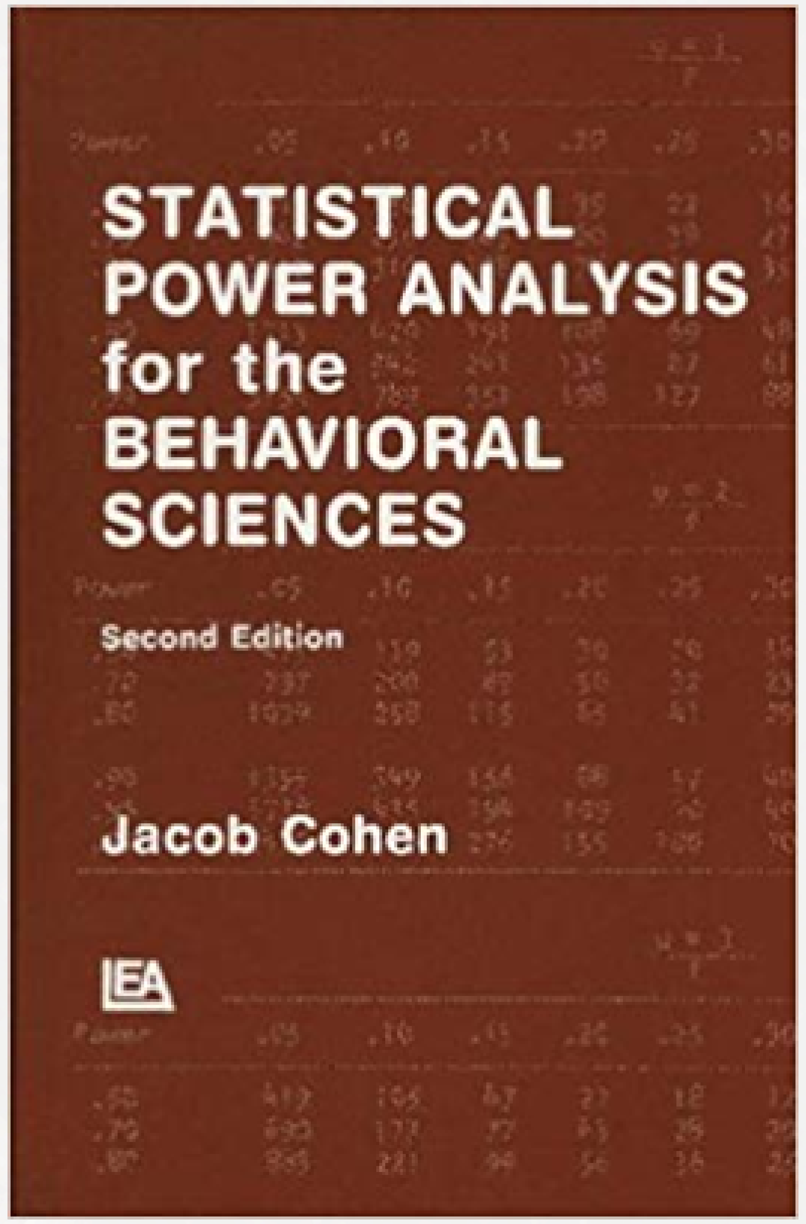
associated with the standard Bonferroni procedure is a substantial reduction in the statistical power of rejecting an incorrect H_o in each test (e.g., Holm, 1979; Perneger, 1998; Rice, 1989). The sequential Bonferroni procedure also incurs reduction in power, but to a lesser extent (which is the reason that the sequential procedure is used in preference by some researchers; Moran, 2003). Thus, both procedures exacerbate the existing problem of low power, identified by Jennions and

Literature


STATISTICAL
POWER ANALYSIS
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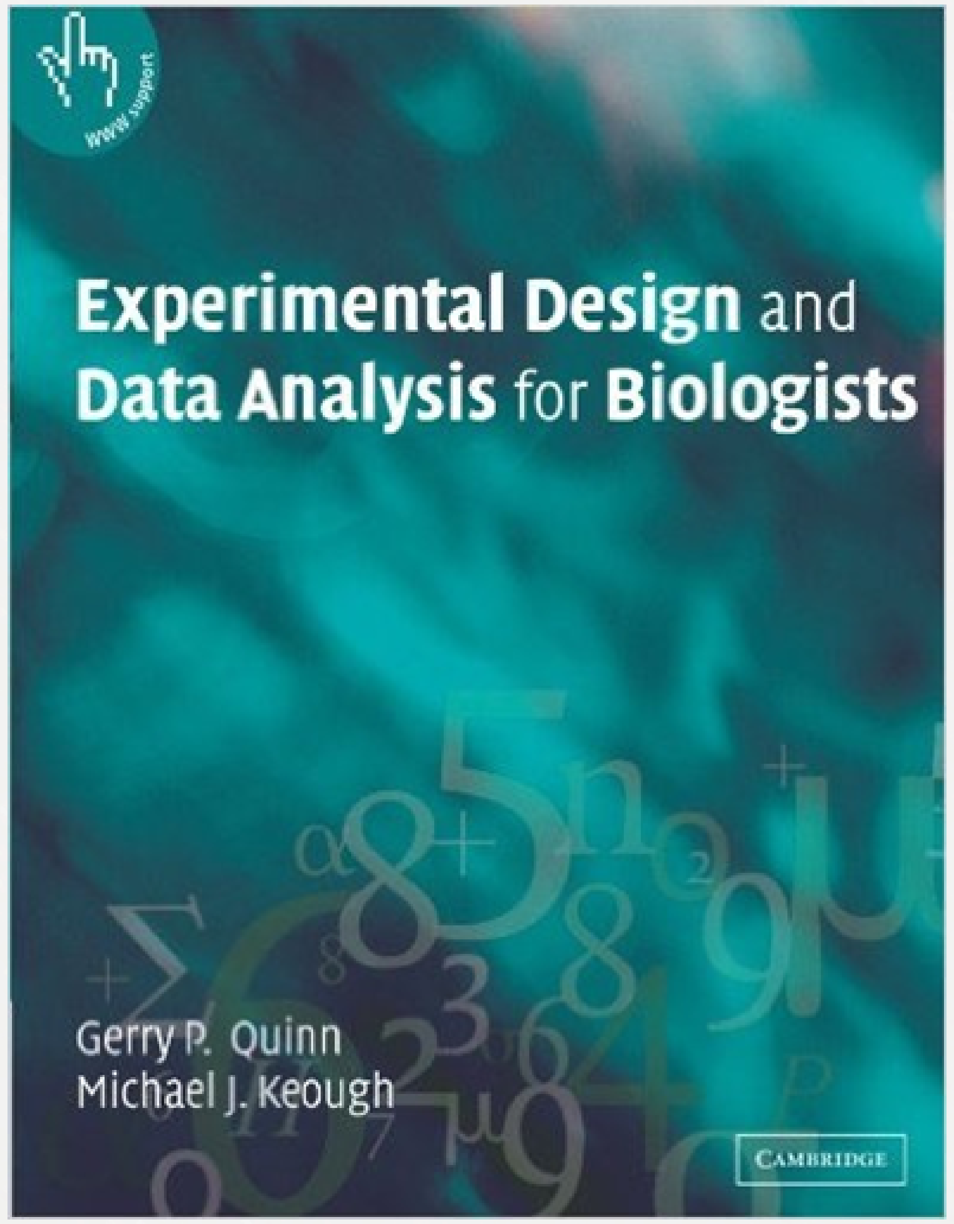
SCIENCES
Second Edition
Jacob Cohen
LEA


WWW support
Experimental Design and
Data Analysis for Biologists
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Code time